

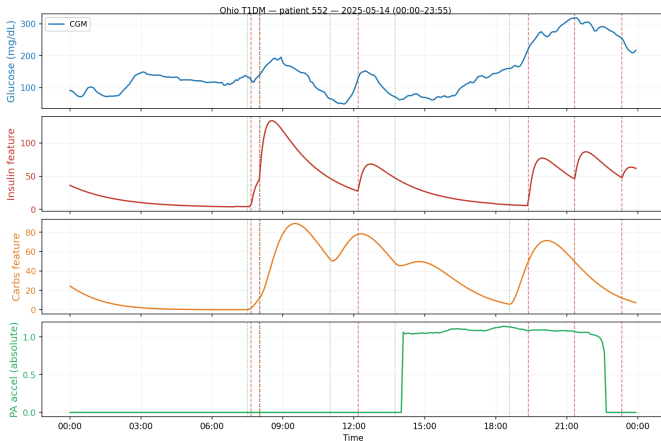
Aligning model behavior with domain expectations via interventional audits

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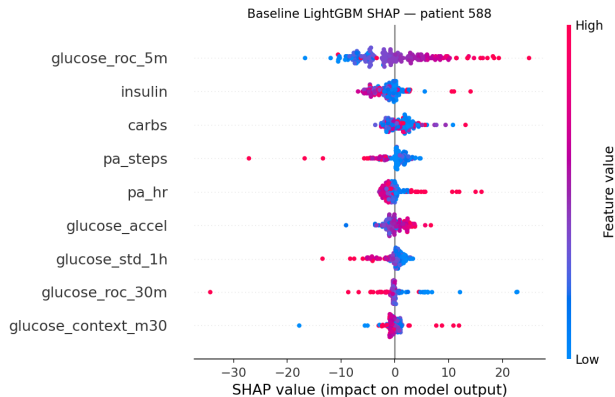
Case study: Ohio T1DM dataset

- Features: insulin impact curve, carbs impact curve, Physical activity sensors, glucose dynamics
- Target: future glucose change at 30 minutes
- Models may mislearn insulin effects when **meals and boluses co-occur** — carbs and insulin features rise together, confounding the learned insulin–glucose association.



The problem: accurate predictions but wrong feature attributions

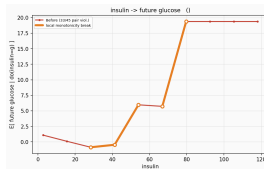
- Predictive models can achieve **low RMSE** yet encode **wrong or inconsistent feature attributions** contradictory to domain expectations.
- **Baseline SHAP** values can suggest but not quantify these inconsistencies.
- Clinical expectation: more insulin should **lower** future glucose.
- SHAP alone does not test that counterfactual; we need an **interventional audit**



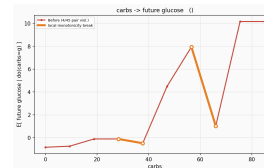
SHAP beeswarm — Ohio T1DM patient 588, baseline LightGBM

Interventional response curves (Partial dependence plots)

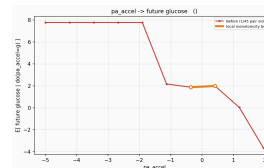
- **Interventional response curve (IRC):** For each level g of a feature X , we perform an intervention $\text{do}(x = g)$ (holding all other features fixed) and compute the mean model prediction $\hat{y}$ for the target variable.
- Clinical expectations: monotonic relationships between insulin, carbs, and activity and the target variable (future blood glucose). insulin ↘, carbs ↗, activity ↘.
- Actual curves: we see violations of these expectations in the curves as indicated by the orange lines.



Insulin — p. 588; ↘ expected (33/45)



Carbs — p. 591; ↗ expected (4/45)

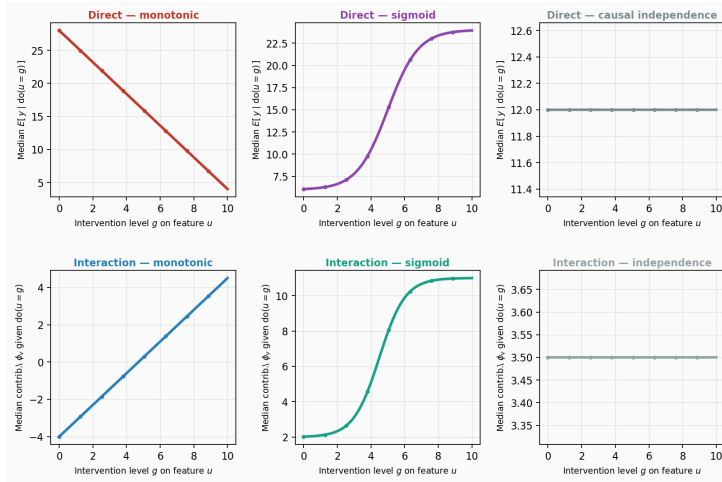


Accel. — p. 567; ↘ expected (1/45)

What can we derive from Interventional response curves?

- Direct feature contributions (upper panels) and feature interactions (lower panels)

- Function characteristics:
Monotonicity (left panels), specific functions, e.g. sigmoid or sinusoidal (middle panels), causal independence (right panels)

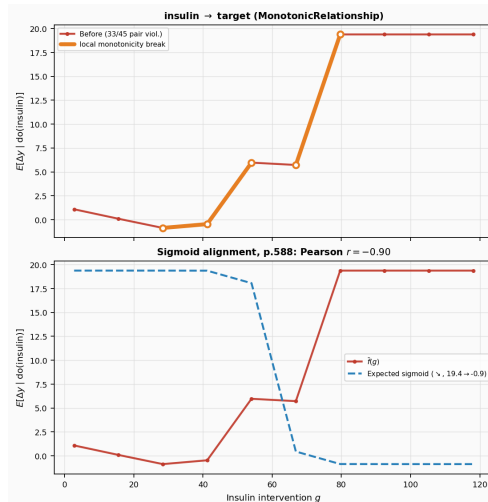


Evaluating the alignment of the interventional response curves with expected behavior

Violations of Monotonicity

Goodness of fit of the interventional response curve to the function modeling expected behavior

We can define expected behavior with a Structural Causal Model (SCM) with additional instructions on edge characteristics.



Approaches for aligning model behavior with domain expectations

Approach	Studies
Model learning: encode SCM monotonicity signs as constraints during training (monotone tree splits, constrained optimization, or penalty terms).	Wolber (2025) ; Koklev (2025); Xie & Tummala (2025); Borghesi et al. (2020)
Model architecture: bake domain structure into the network—e.g. physiological LSTM with fixed insulin/CHO preprocessing layers that separate time-action profiles before sequence modeling.	Prendin et al. (2023)
Preprocessing: residualize insulin against glucose-context features so bolus signal is orthogonal to confounding CGM dynamics before any learner.	Akinwumi et al. (2026)
Feature engineering: add explicit insulin-context and carbs-context interaction terms to expose meal-bolus co-occurrence in the input space.	Butt et al. (2023); Beauchamp et al. (2021)
Synthetic augmentation: generate SCM-aligned synthetic rows.	Noguer et al. (2022); Prendin & Facchinetti (2025); Kalita et al. (2024)
Latent confounder: impute latent confounder feature(s) from feature interactions	Sun et al. (2020–2023); Bing et al. (2022); Prendin et al. (2023)

What our approach aims to do differently

IRAlign = align tabular models with domain rules using interventional response curves.

- **No accuracy compromise** — RMSE/feasibility trade-off (e.g. on Pareto front).
- **Model-agnostic** — wraps any tabular model.
- **Fast IRC scoring**
- **Flexible rules** — monotonic, sigmoid, flat, etc; implementable not only for Diabetes modeling

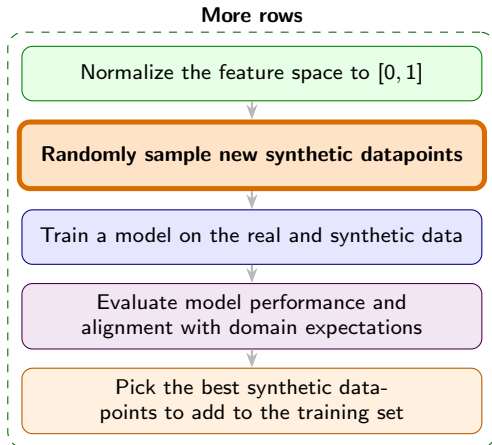
Under investigation

We currently investigate two approaches: Synthetic augmentation vs. latent confounder imputation.

Approach A: Synthetic data augmentation

Idea

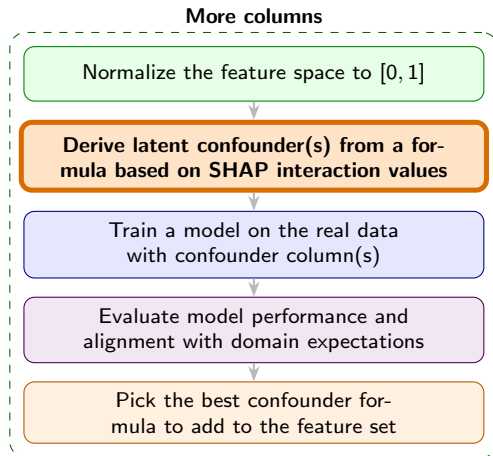
Add synthetic training rows the patient never logged — e.g. insulin boluses in the absence of meals. A diabetic patient would rarely do this. However, these datapoints are needed to learn correct insulin behavior.



Approach B: Latent confounder imputation

Idea

Add a latent confounder feature — a new column from a formula over SHAP interaction values — to represent unobserved drivers the sensors never record: e.g. meal overestimation, circadian rhythm, absorption variability, or combined interaction effects that distort the insulin–glucose link.



Preliminary Head-to-head: all Ohio T1D patients (insulin monotonicity (1=no violation of monotonicity))

	Val RMSE	Test RMSE	Insulin feasibility	Runtime
Baseline	22.44 ± 3.62	21.19 ± 3.00	0.79 ± 0.28	0.1 ± 0.0 s
Synthetic augmentation	22.38 ± 3.55	21.21 ± 3.00	0.85 ± 0.24	135.8 ± 20.8 s
Latent confounder	22.05 ± 3.61	21.91 ± 3.07	0.85 ± 0.26	68.9 ± 6.5 s

